Exercise for Sect. 2.1

Exercise 2.1.1

Show that the space $(\mathbb{T}, \mathcal{B}_{\mathbb{T}}, m_{\mathbb{T}})$ is isomorphic as a measure space to $(\mathbb{T}^2, \mathcal{B}_{\mathbb{T}^2}, m_{\mathbb{T}^2})$.

Proof: For any $x \in \mathbb{T}$, express x in an infinite series $(x_i)_{i\in\mathbb{N}}$ by x_i is the i-th binary place of x. Let $D = \left\{\frac{a}{2^k} : a, k \in \mathbb{N} \text{ and } a < 2^k\right\}$ denote the set of dyadic rationals and D is measure zero since D is countable. Then, define $A^2 \subset \mathbb{T}^2$ by $A^2 = T^2 \setminus (\mathbb{T} \times D \cup D \times \mathbb{T})$. A^2 is measure 1 since $D \times \mathbb{T}$ and $\mathbb{T} \times D$ are measure zero. In this construction, A^2 is the set only containing the points have infinitely many 0 and 1 digits. Define $f : 0.x_1x_2x_3\cdots \mapsto (0.x_1x_3x_5\cdots, 0.x_2x_4x_6\cdots)$ and $A = f^{-1}(A^2)$. A is measurable and measure 1 since $A = \mathbb{T} \setminus (D \cup D_{\text{odd}} \cup D_{\text{even}})$ where the two sets stand for the set of points contiains fintely many 0 or 1 in odd or even digits.

To prove f is bimeasurable, define the cylender set $I_{x_1,x_2,\ldots,x_n} = \{y \in A : y_i = x_i \text{ for } i \le n\}$ and collection of the intervals $\mathcal{S} := \left\{I_{x_1,\ldots,x_n} : x_i \in \{0,1\} \text{ and } n \in \mathbb{N}\right\}$ which is a π-system on A that $\sigma(\mathcal{S}) = \mathcal{B}_{\mathbb{T}}$. Similarly, $\mathcal{S}^2 := \left\{R_{(x_1,\ldots,x_n),(y_1,\ldots,y_m)}\right\}$ where $R_{(x_1,\ldots,x_n),(y_1,\ldots,y_m)} = \left\{(z,w) \in A^2 : z_i = x_i, w_j = y_j \text{ for } i \le n \text{ and } j \le m\right\}$ and $\sigma(\mathcal{S}^2) = \mathcal{B}_{\mathbb{T}^2}$. Then, we want to show f and f^{-1} maps the cylinder sets into measurable sets.

Given an $I_{x_1,x_2,\ldots,x_{2k}}$, $f\left(I_{\ldots,x_{2k}}\right) = R_{(x_1,\ldots,x_{2k-1}),(x_2,\ldots,x_{2k})}$, and given $I_{x_1,x_2,\ldots,x_{2k+1}}$ get $f(I) = R_{(\ldots,x_{2k+1}),(\ldots,x_2k)}$. These two situation is trivially measurable.

In contrast, for given $R_{(\ldots,x_n),(\ldots,y_m)}$, we can divide it into finite disjoint union of cylinders $R_{(\ldots,x_k),(\ldots,y_k)}$ where $k \le \max(n,m)$. Then the preimage is easy to calculate, $f^{-1}\left(R_{(\ldots,x_k),(\ldots,y_k)}\right) = I_{x_1,y_1,\ldots,x_k,y_k}$ measurable.

Last is to check if f preserves measure. $m_{\mathbb{T}}\left(I_{\ldots,x_k}\right) = 2^{-k}$ and $m_{\mathbb{T}^2}\left(R_{(\ldots,x_n),(\ldots,y_m)}\right) = 2^{-n} \times 2^{-m} = 2^{-nm}$. From the result of mapping above, f and f^{-1} preserves measure on cylinder sets and thus preserves on all measurable sets. Thus, $(\mathbb{T}, \mathcal{B}_{\mathbb{T}}, m_{\mathbb{T}})$ is isomorphic as a measure space to $(\mathbb{T}^2, \mathcal{B}_{\mathbb{T}^2}, m_{\mathbb{T}^2})$. $\square$

Exercise **2.1.2**

Show that the measure-preserving system $(\mathbb{T}, \mathcal{B}_{\mathbb{T}}, m_{\mathbb{T}}, T_4)$, where $T_4(x) = 4x (\mathrm{mod}\ 1)$, is measurably isomorphic to the product system $(\mathbb{T}^2, \mathcal{B}_{\mathbb{T}^2}, m_{\mathbb{T}^2}, T_2 \times T_2)$.

Proof: By the construction above, this is trivial by treating T_k as moving all digit leftward k times. □

Exercise **2.1.3**

For a map $T : X \to X$ and sets $A, B \subseteq X$, prove the following.

- $\chi_A(T(x)) = \chi_{T^{-1}(A)}(x)$;
- $T^{-1}(A \cap B) = T^{-1}(A) \cap T^{-1}(B)$;
- $T^{-1}(A \cup B) = T^{-1}(A) \cup T^{-1}(B)$;
- $T^{-1}(A \Delta B) = T^{-1}(A) \Delta T^{-1}(B)$.

Which of thexe properties also hold with the pre-image under T^{-1} replaced by the forward image under T?

Proof: The four are easy and after replacing, only union still holds. □

Exercise **2.1.4**

Let X be a compact abelian group and let $T : X \to X$ be a countinuous homomorphism. Does T still preserve the Haar measure m_X on X?

Proof: Let T be the constant map from X to identity of X, T satisfies the assumption and may not preserve measure if X contains more than one elements. $\square$

Exercise **2.1.5**

1. Find a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with a non-trivial factor map $\varphi : X \to X$.
2. Find an invertible measure-preserving system $(X, \mathcal{B}, \mu, T)$ with a non-trivial factor map $\varphi : X \to X$.

Proof:

1. $X = \{0, 1\}^{\mathbb{N}}$ be the sample space of a infinite series of outcome of Bernoulli trials, and $\mathcal{B}$ is the σ-algebra generate by all cylinder sets. μ is the Bernulli measure with probability $\frac{1}{2}$ and $\frac{1}{2}$. Define T by $(Tx)_n = x_{n+1}$ for $x \in X$ which is the left shift map on X. Then, claim that $\varphi(x_n) = x_n + x_{n+1} \bmod 2$ is a non-trivial factor map on X to itself.

   First, φ is not an isomorphism since it maps two elements (one solution and its comjugate) to one element, and φ is surjective since for $y \in X, y = y + 0 \bmod 2$. Then, $(\varphi \circ T(x))_n = \varphi\big((Tx)_n\big) = \varphi(x_{n+1}) = x_{n+1} + x_{n+2} \bmod 2 = (\varphi(x))_{n+1} = T\big((\varphi(x))_n\big) = (T \circ \varphi(x))_n$.

   Last, for C is an cylinder set given by $\left\{x \in X : x_{n_1} = c_1, \ldots, x_{n_k} = c_k\right\}$. If n_i and n_j are not next to each other, the choice of each one won't affect to another one, so we can divides into not adjacency parts and multiply them after computing each part. Thus, we only focus on the situation that $n_j = n_i + 1$, $n_l = n_j + 1$ and so on. If $k = 1$, $\mu\big(\varphi^{-1}(C)\big) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2} = \mu(C)$(11 and 00 or 01 and 10.) If $k = 2$, $\mu\big(\varphi^{-1}(C)\big) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot 2 = \frac{1}{4} = \mu(C)$ (the whole process actually depends on the first one.) In conclusion, $\mu\big(\varphi^{-1}(C)\big) = \mu(C)$ for all cylinder sets and thus φ is measure-preserving on $\mathcal{B}$.
2. Just modify X into $\{0, 1\}^{\mathbb{Z}}$, the other settings are still the same.

$\square$

Exercise **2.1.6**

Prove that the circle rotation $R_\alpha : \mathbb{T} \to \mathbb{T}, R_\alpha(t) = t + a_\alpha (\text{mod } 1)$ is not measurably isomorphic to the circle-doubling map $T_2 : \mathbb{T} \to \mathbb{T}, T_2(t) = 2t(\text{mod } 1)$.

Proof: Assume there is an isomorphic measurable function $\varphi : \mathbb{T} \to \mathbb{T}$ such that $R_\alpha \circ \varphi = \varphi \circ T_2$ almost everywhere. Then, for almost every $x \in \left(0, \frac{1}{2}\right)$ satisfies the equation,

$$\varphi(x) + \alpha \text{ mod } 1 = \varphi(2x)$$

$$(\varphi(2x) - \varphi(x)) \text{ mod } 1 = \alpha$$

$$\Rightarrow \begin{cases} \varphi(2x) - \varphi(a) = \alpha \\ \varphi(2x) - \varphi(b) = \alpha - 1' \end{cases}$$

where one of a, b is x and another is $x + \frac{1}{2}$. Minus that two equations can get $\varphi(b) - \varphi(a) = 1$; however, it is impossible since range of φ is $[0, 1)$. Thus, the two map are not measurable isomorphic. $\square$

Exercise **2.1.7**

Let $\mathfrak{X} = (X, \mathcal{B}, \mu, T)$ be any measure-preserving system. A sub-σ-algebra $\mathcal{A} \subseteq \mathcal{B}_X$ with $T^{-1}\mathcal{A} = \mathcal{A}$ modulo μ is called a *T-invarient sub-σ-algebra*. Show that the system $\tilde{\mathfrak{X}} = \left(\tilde{X}, \tilde{\mathcal{B}}, \tilde{\mu}, \tilde{T}\right)$ defined by

- $\tilde{X} = \left\{x \in X^{\mathbb{Z}} : x_{k+1} = T\left(x_k\right) \text{ for all } k \in \mathbb{Z}\right\}$;
- $\left(\tilde{T}(x)\right)_k = x_{k+1}$ for all $k \in \mathbb{Z}$ and $x \in \tilde{X}$;
- $\tilde{\mu}\left(\left\{x \in \tilde{X} : x_0 \in A\right\}\right) = \mu(A)$ for any $A \in \mathcal{B}$, and $\tilde{\mu}$ is invariant under $\tilde{T}$;
- $\tilde{\mathcal{B}}$ is the smallest $\tilde{T}$-invariant σ-algebra for which the map $\pi : x \mapsto x_0$ from $\tilde{X}$ to X is measurable;

is an invertible measure-preserving system, and that the map $\pi : x \mapsto x_0$ is a factor map. The system $\tilde{\mathfrak{X}}$ is called the invertible extension of $\mathfrak{X}$.

Proof:

- Invertible: $\left(\tilde{T}^{-1}(x)\right)_k = x_{k-1}$ is clearly the inverse map, and $\tilde{T}^{-1}$ is also measurable.
- Measure-preserving: $\tilde{\mu}$ is invarant under $\tilde{T}$ implies the measure-preserving condition.
- Factor map: for any $x \in \tilde{X}$, $\pi \circ \tilde{T}(x) = x_1 = T\left(x_0\right) = T(\pi(x))$.

$\square$

Exercise **2.1.8**

Show that the invertible extension $\tilde{\mathfrak{X}}$ of a measure-preserving system $\mathfrak{X}$ constructed above has the following universal property. For any extension

$$\varphi : (Y, \mathcal{B}_Y, \nu, S) \to (X, \mathcal{B}_X, \mu, T)$$

for which S is invertible, there exists a unique map

$$\tilde{\varphi} : (Y, \mathcal{B}_Y, \nu, S) \to \left(\tilde{X}, \tilde{\mathcal{B}}, \tilde{\mu}, \tilde{T}\right)$$

for which $\varphi = \pi \circ \tilde{\varphi}$.

Proof:

□

Exercise **2.1.9**

1. Show that the invertible extension of the circle-doubling map T_2,

$$X_2 = \{x \in \mathbb{T}^{\mathbb{Z}} : x_{k+1} = T_2 x_k \text{ for all } k \in \mathbb{Z}\},$$

is a compact abelian group with respect to the coordinate-wise addition defined by $(x+y)_k = x_k + y_k$ for all $k \in \mathbb{Z}$, and the topology inherited from the product topology on $\mathbb{T}^{\mathbb{Z}}$.

2. Show that the diagonal embedding $\delta(r) = (r, r)$ embeds $\mathbb{Z}\left[\frac{1}{2}\right]$ as a descrete subgroup of $\mathbb{R} \times \mathbb{Q}_2$, and that $X_2 \cong \mathbb{R} \times \mathbb{Q}_2 / \delta\left(\mathbb{Z}\left[\frac{1}{2}\right]\right) \cong \mathbb{R} \times \mathbb{Z}_2 / \delta(\mathbb{Z})$ as compact abelian groups. In particular, the map $\tilde{T}_2$ (which mau be thought of as the left shift on X_2, or as the map that doubles in each coordinate) is conjugate to the map

$$(s, r) + \delta\left(\mathbb{Z}\left[\frac{1}{2}\right]\right) \mapsto (2s, 2r) + \delta\left(\mathbb{Z}\left[\frac{1}{2}\right]\right)$$

on $\mathbb{R} \times \mathbb{Q}_2 / \delta\left(\mathbb{Z}\left[\frac{1}{2}\right]\right)$. The group X_2 constructed in this exercise is a simple example of a *solenoid*.

Proof:

□